

Concurrent Programming - Exercises 4

Isomorphism, Trace Equivalence and (Strong) Bisimulation

Consult Chap. 3-1-3,3 e 3.5 of [1].

1. Let

$$\begin{aligned} P &:= a!.0 + b!.0 \\ Q &:= b!.0 + a!.0 \end{aligned}$$

determine if the following relations hold

$$\begin{aligned} P &\sim_{tr} Q \\ a!.P &\sim_{tr} a!.Q \\ a!.P + a!.P &\sim_{tr} a!.Q + a!.Q \end{aligned}$$

and also for $\sim_{iso}$?

2. Show $(b!.0 + c!.0) \sim_{iso} (c!.0 + b!.0)$ but $a!.(b!.0 + c!.0) + a!.(b!.0 + c!.0) \not\sim_{iso} a!.(b!.0 + c!.0) + a!.(c!.0 + b!.0)$ and conclude that $\sim_{iso}$ is not a congruence.
3. Show that for every $P, Q, R \in CCS$

$$\begin{aligned} P + Q &\sim_{tr} Q + P \\ (P + Q) + R &\sim_{tr} P + (Q + R) \\ P + 0 &\sim_{tr} P, \end{aligned}$$

and $\alpha.(P + Q) \sim_{tr} \alpha.P + \alpha.Q$.

4. Let

$$\begin{aligned} CTM &:= coin?.(coffee!.CTM + tea!.CTM) \\ CTM' &:= coin?.coffee!.CTM' + coin?.tea!.CTM' \end{aligned}$$

Show that $CTM \sim_{tr} CTM'$.

5. Solve bisimulation problems in <http://tinyurl.com/pseuco>
6. Solve (strong) bisimulation problems in PseuCo Book <https://book.pseuco.com/#/read/equality/workout>
7. Show that
 - (a) $\sim$ is an equivalence relation.
 - (b) $\sim$ is the largest bisimulation.
 - (c) $s \sim t$ iff for each $\alpha \in Act$

- If $s \xrightarrow{\alpha} s'$ then t' such that $t \xrightarrow{\alpha} t'$ e $s' \sim t'$
- If $t \xrightarrow{\alpha} t'$ then s' such that $s \xrightarrow{\alpha} s'$ and $s' \sim t'$.

8. Let $TS = (S, Act, \longrightarrow)$ such that $S = \{s_i \mid i \geq 1\} \cup \{t\}$, $Act = \{a\}$ e $\xrightarrow{a} = \{(s_i, s_{i+1}) \mid i \geq 1\} \cup \{(t, t)\}$.

Show that $s_1 \sim t$ proving that $R = \{(s_i, t) \mid i \geq 1\}$ is a bisimulation.

9. Let P and Q be defined by

$$P := a.P_1 + b.P_2$$

$$P_1 := c.P$$

$$P_2 := c.P$$

and

$$Q := a.Q_1 + b.Q_2$$

$$Q_1 := c.Q_3$$

$$Q_2 := c.Q_3$$

$$Q_3 := a.Q_1 + b.Q_2$$

Show that $P \sim Q$ presenting a bisimulation that contains (P, Q) . Draw their LTSs and test in pseuCo.com.

10. Let P and Q be defined by

$$P := a.P_1$$

$$P_1 := b.P + c.P$$

and

$$Q := a.Q_1$$

$$Q_1 := b.Q_2 + c.Q$$

$$Q_2 := a.Q_3$$

$$Q_3 := b.Q + c.Q_2$$

Show that $P \sim Q$ presenting a bisimulation that contains (P, Q) .

11. Show that if $P \sim Q$ and $\alpha \in Act$, $R \in CCS$ and $H \subseteq Com$, then

$$\alpha.P \sim \alpha.Q$$

$$P + R \sim Q + R$$

$$R + P \sim R + Q$$

$$P|R \sim Q|R$$

$$R|P \sim R|Q$$

$$P \setminus H \sim Q \setminus H$$

12. Let

$$\begin{aligned} P &:= a.(b.0 + c.0) \\ Q &:= a.b.0 + a.c.0 \end{aligned}$$

Show that P and Q are not strongly bisimilar.

13. Show that

$$\begin{aligned} P + Q &\sim Q + P \\ P + 0 &\sim P \\ (P + Q) + R &\sim P + (Q + R) \end{aligned}$$

14. Show that for all $P, Q, R \in CCS$,

$$\begin{aligned} P|Q &\sim Q|P \\ P|0 &\sim P \\ (P|Q)|R &\sim P|(Q|R) \end{aligned}$$

Show that these are bisimulations

$$\begin{aligned} &\{(P|Q, Q|P) \mid P, Q \in CCS\}, \\ &\{(P|0, P) \mid P \in CCS\}, \\ &\{((P|Q)|R, P|(Q|R)) \mid P, Q, R \in CCS\}. \end{aligned}$$

15. Let:

$$\begin{aligned} C_0 &:= inc.C_1 \\ C_n &:= inc.C_{n+1} + dec.C_{n-1}, \text{ para } n \geq 1 \end{aligned}$$

and $C := inc.(C|dec.0)$.

- (a) Draw initial fragments of $\llbracket C \rrbracket_\Gamma$ and $\llbracket C_0 \rrbracket_\Gamma$.
- (b) Show that the following relation is a bisimulation.

$$\begin{aligned} \mathcal{R} = & \{(C | \prod_{i=1}^k P_i, C_n) \mid k \geq 0 \wedge (P_i = 0 \vee P_i = dec.0) \\ & \wedge \text{the number of } i \text{ with } P_i = dec.0 \text{ is } n\} \end{aligned}$$

Consider $(C | \prod_{i=1}^k P_i, C_n) \in \mathcal{R}$.

Show that

1. if $C | \prod_{i=1}^k P_i \xrightarrow{\alpha} P$ exists Q such that $C_n \xrightarrow{\alpha} Q$ and $(P, Q) \in \mathcal{R}$.
2. if $C_n \xrightarrow{\alpha} Q$ exists P such that $C | \prod_{i=1}^k P_i \xrightarrow{\alpha} P$ and $(P, Q) \in \mathcal{R}$.

16. Consider a 1-Buffer.

$$B \quad := \quad put?.get?.B$$

For $n \geq 1$ we can consider a Buffer with capacity n , where B_i^n is a buffer with capacity n with $0 \leq i \leq n$ elements.

$$\begin{aligned} B_0^n &:= put?.B_1^n \\ B_i^n &:= put?.B_{i+1}^n + get?.B_{i-1}^n, \quad 0 < i < n \\ B_n^n &:= get?.B_{n-1}^n \end{aligned}$$

- (a) Verify that $B \sim_{iso} B_0^1$ (draw their LTSs).
- (b) Verify that $B_0^2 \sim B_0^1 | B_0^1$ (draw their LTSs).
- (c) Show that for $n \geq 1$,

$$B_0^n \sim \underbrace{B_0^1 | B_0^1 | \dots | B_0^1}_n.$$

showing that the following relation is a bisimulation:

$$\mathcal{R} = \{(B_i^n, B_{i_1}^1 | B_{i_2}^1 | \dots | B_{i_n}^1) \mid i_j \in \{0, 1\} \wedge \sum_{j=1}^n i_j = i\}$$

References

- [1] L. Aceto, A. Ingólfssdóttir, K.G. Larsen, and J. Srba. *Reactive Systems: Modelling, Specification and Verification*. Cambridge University Press, 2007.